



Monthly monitoring of urban development and renewal at the block-level in China using Sentinel-2 time series[☆]

Haixu He^{a,b}, Jining Yan^{b,c}, Lirong Liu^d, Xu Long^b, Runyu Fan^b, Zhongchang Sun^e

^a School of Future Technology, China University of Geosciences, Wuhan 430074, China

^b School of Computer Science, China University of Geosciences, Wuhan 430074, China

^c Engineering Research Center of Natural Resource Information Management and Digital Twin Engineering Software, Ministry of Education, Wuhan 430074, China

^d Land Satellite Remote Sensing Application Center, MNR, Beijing 100048, China

^e Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China

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ABSTRACT

Urban renewal has been elevated to a national strategy in China, leading to rapid development and transformation of street blocks. However, monitoring construction events at high temporal resolution remains challenging due to the limitations of existing methods, which often struggle with noise interference and lack continuous monitoring capabilities. To address this, we propose Semantic Similarity Contrast-based Street Block Monitoring (SSC-SB), a method that leverages Sentinel-2 time series imagery for automated, high-frequency detection of street block development and renewal. By extracting deep semantic features with a pretrained encoder, SSC-SB analyzes similarity curves to identify development and demolition construction events. Applied to the Middle Yangtze River Basin (MYRB) urban agglomeration shows that SSC-SB achieves 90.4% spatial domain accuracy, with construction start and end date detection accuracies of 68.8% and 54.9%, respectively. Results indicate an increasing emphasis on urban renewal, as demolished street blocks outnumbered new developments for the first time in 2023, with Hunan Province leading in renewal efforts, where renewal blocks accounted for 41.5% of all changed street blocks, reflecting a balanced focus on expansion and infrastructure renewal. Transfer experiments in Xi'an further demonstrate that SSC-SB retains up to 80% of the performance of a locally trained model when applied across regions without fine-tuning, indicating a decent level of generalizability. By providing fine-grained, continuous monitoring, SSC-SB presents a scalable solution for tracking urban transformation.

1. Introduction

“Urban Renewal” was officially included in China’s government work report in 2021, marking its elevation to a national strategy (State Council, 2021). Urban renewal refers to the process of reconfiguring urban space resources through maintenance, renovation, demolition, and other methods as urbanization approaches maturity, in order to better meet public needs and adapt to economic development. Currently, over 400 cities in China have established urban renewal leadership groups. In the first half of 2024 alone, 33,000 urban renewal projects for old residential neighborhoods were launched nationwide (Central People’s Government, 2024).

Urban development typically refers to the expansion phase of cities, characterized by the outward growth of urban areas through the addition of new construction land. Its focus lies in incremental development, aiming to meet the demands of rapid urbanization for housing, infrastructure, and public services (Angel et al., 2005; Hongsheng et al., 2016). In contrast, urban renewal pertains to the optimization of existing urban space in the context of mature urbanization. It involves the reuse, renovation, and functional transformation of built-up areas such as old neighborhoods and underutilized land, aiming to enhance spatial efficiency and improve residents’ quality of life in response to evolving socio-economic needs (Chen et al., 2019; Xiao et al., 2024; Zhao et al., 2024). Thus, efficiently monitoring and evaluating urban development has become a critical topic in urban studies and policy formulation (Wang et al., 2022; Jing et al., 2021; Zhang et al., 2024).

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* Corresponding author at: School of Computer Science, China University of Geosciences, Wuhan 430074, China.

E-mail address: yanjn@cug.edu.cn (J. Yan).

Accurate monitoring of urban construction enables governments to identify and locate development hotspots promptly, providing scientific support for rational resource allocation, environmental protection, and the establishment of new urban development frameworks. This can help mitigate the negative impacts of urbanization and promote sustainable urban development (Al-Raei, 2024; Aslam et al., 2023; Li et al., 2023a).

As the fundamental unit of urban management, the street block is a key focus for studying urban functional layouts and development planning (Jing et al., 2021; Jendryke et al., 2017; Huang et al., 2017). Research on street blocks, including land use classification, building density evaluation, building height inversion, and disaster recovery potential assessment, provides critical support for the digitalization of smart cities (Zhang et al., 2017; Li et al., 2020; Goldblatt et al., 2020). Remote sensing technology, with its large-scale coverage and precise data, has become a core tool for these studies. In particular, time-series remote sensing imagery, combined with high spatial and temporal resolution, effectively captures dynamic changes at the street block level. This is especially critical for urban management in the context of rapid urban expansion and old town renewal. The real-time and continuous monitoring capabilities of remote sensing data are essential for addressing these challenges (Wang et al., 2022; He et al., 2024). Building on this, the study explores the use of remote sensing technology for dynamic monitoring of street block construction. The goal is to assist decision-makers in quickly identifying development hotspots and accurately assessing construction progress.

Currently, two main approaches are used for monitoring construction of street blocks based on remote sensing time series: land cover classification-based methods and feature similarity-based methods. Land cover classification-based methods first perform annual land cover classification and then analyze the proportions of different land cover types at the street block scale to detect construction time and type (Jing et al., 2021). While this method can effectively identify construction date, type, and magnitude, its primary limitation lies in its scalability to higher temporal resolutions (e.g., monthly-scale detection). High-frequency land cover classification is constrained by the prohibitive cost of implementation (Wang et al., 2022; He et al., 2024). To address this, some researchers have developed feature similarity-based methods for street block construction detection. These methods utilize monthly remote sensing imagery of street blocks, extracting monthly texture features using Histogram of Oriented Gradient (HOG) operators and clustering high-similarity images to detect construction events (Wang et al., 2022). Although this approach is simple and easy to use, it is highly susceptible to noise (Huang et al., 2020; Zhu et al., 2022), and the extracted features often fail to meet the accuracy requirements for construction monitoring (Lyu et al., 2024). Additionally, the pairwise contrast of features across time points prevents continuous monitoring of construction events. When new observations are added, the detection results must be readjusted, limiting the method's ability to accurately capture dynamic construction and perform long-term monitoring during rapid urbanization.

Other approaches include change detection using bi-temporal street block or building imagery (Jendryke et al., 2017; Huang et al., 2017; Ismail and Awad, 2022; Zhang et al., 2023; Xiong et al., 2024), which lack sufficient temporal information to represent the full construction process. Pixel-level change detection has also been employed to track street block construction (Zhu and Woodcock, 2014; Suh et al., 2024; Li et al., 2023b; Zhu et al., 2020). While these methods provide finer-grained representations, it does not align with our focus on street blocks and incurs high computational costs, so it will not be discussed further. In summary, existing methods lack the ability to directly monitor street block construction with both high temporal resolution and accuracy. Moreover, there is a need for a robust technique capable of tracking and precise monitoring under the challenges of rapid urbanization and complex environments.

To meet the demand for high-frequency monitoring under rapid urban renewal, this study proposes a Semantic Similarity Contrast-based Street Block construction monitoring (SSC-SB) method. The method first uses a supervised contrastive learning model to extract semantic features from remote sensing images at the street block level. It then builds a continuous temporal comparison mechanism to evaluate changes in semantic features across time. Finally, a change criterion is defined to determine whether a construction event has occurred. The contributions of SSC-SB are reflected in:

- Leverages contrastive learning to extract temporally consistent and semantically discriminative features, enhancing robustness to seasonal and cloud-related noise.
- Focuses on modeling the continuity and gradual changes of street block development, enabling effective tracking of subtle use type shifts over time.
- Supports monthly street block-level monitoring, meeting the demand for high-frequency tracking of urban construction events.

2. Study area and material

We validated the effectiveness of the proposed method using the middle Yangtze River basin (MYRB) urban agglomeration as a case study. MYRB is one of the most economically dynamic regions in China (Fig. 1(a)) and serves as a key pilot area for national initiatives in new urbanization and urban renewal (Wang and Wang, 2025). This region spans parts of Hubei, Hunan, and Jiangxi provinces, encompassing the core cities of Wuhan, Changsha, and Nanchang. In recent years, it has experienced rapid urban expansion and industrial restructuring, with frequent street block development and renewal events, providing valuable insights into the spatial patterns of different urban renewal models (Wu et al., 2025).

The study area covers a total of 30 cities, as shown in Fig. 1(b). The main urban boundaries of each city were extracted from the Global Artificial Impervious Area (GAIA) dataset and refined using the Sentinel-2 imagery for December 2024 to ensure complete coverage of street blocks (Gong et al., 2020). Additionally, secondary roads from Open Street Map were manually corrected and topologically validated in ArcGIS based on December 2024 imagery. The resulting closed polygon vectors were used to clip the remote sensing time series imagery (Fig. 1(c)). A total of 24,050 street blocks were extracted across MYRB for further analysis. The distribution of street blocks per city is summarized by province in Fig. 1(d). The capital cities Wuhan, Changsha, and Nanchang each account for nearly 50% of their respective province's street blocks. Therefore, these three cities were selected as representative cases for subsequent analysis.

We obtained Sentinel-2 Level-2A imagery from January 2021 to December 2024 for this region using Google Earth Engine (GEE). Ten bands at 10 m and 20 m resolutions were selected. After removing cloud-covered pixels, monthly median composites were generated, resulting in a time series of 48 images. Spatial gaps in the time series were linearly interpolated.

3. Methods

Street block state changes typically consist of two phases: a stable period followed by a construction event. The stable period refers to a stage during which the street block remains unchanged and experiences no construction events. After remaining stable for some time, a street block may enter construction, during which different trajectories can occur (Xu et al., 2019b; Xu and Akita, 2021). In this study, we categorize street block changes into two types: development and renewal (Fig. 2). Development refers to the change from an undeveloped to a developed state, with the construction event defined as the “development” event subcategory. Renewal, by contrast, occurs on the basis of an already developed street block, which undergoes a “demolition” event

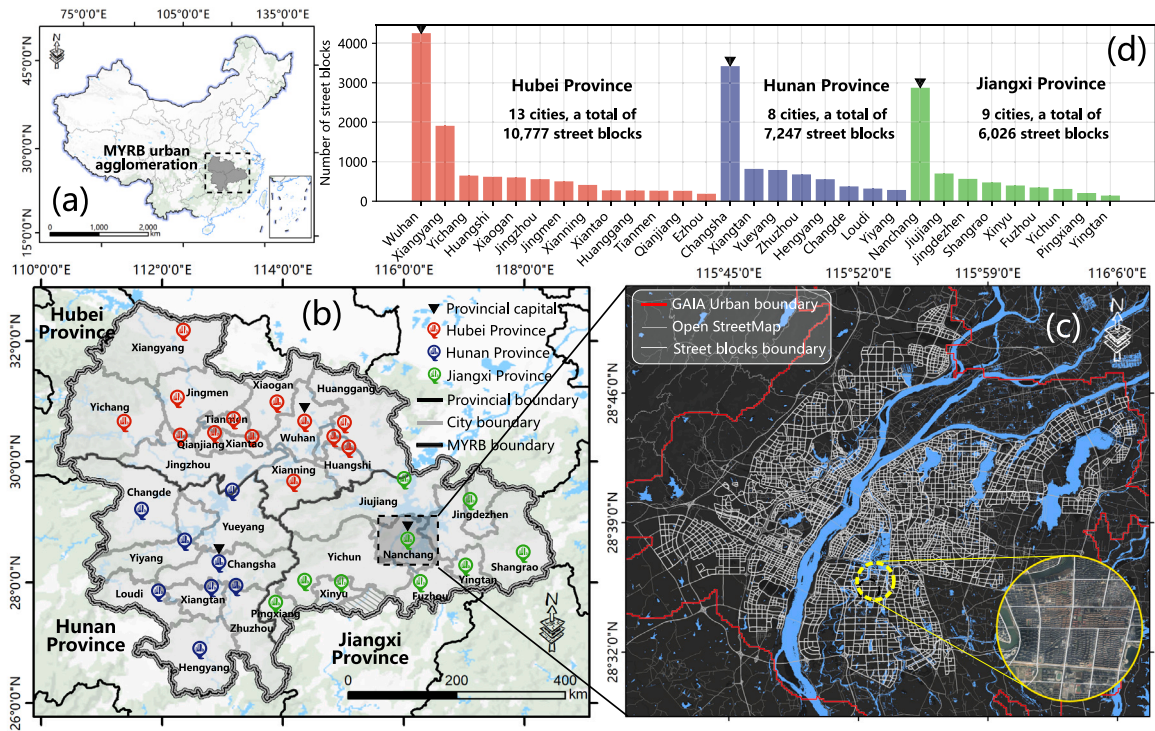


Fig. 1. Basic information of the study area. (a) Location of the middle Yangtze River basin (MYRB) urban agglomeration study area in China. (b) Geographic distribution of the 30 cities in MYRB (excluding Ji'an) and their central urban areas, with three colors representing different provinces. (c) Street blocks delineation using Nanchang as an example. (d) Statistical summary of street block counts across the 30 cities. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

before entering the second development event. Thus, whether a street block has experienced a demolition event serves as the key criterion for distinguishing renewal from development. The aim of SSC-SB is to monitor and classify these construction events. Notably, such events seldom occur instantaneously but instead evolve gradually, exhibiting clear temporal continuity. To capture this evolutionary process, SSC-SB adopts a two-stage strategy, as illustrated in Fig. 3.

First, a semantic encoder is trained to generate robust semantic representations for short-term street block samples, where “short-term” refers to a short temporal window (e.g., a few consecutive months) of remote sensing imagery rather than the entire multi-year sequence. These short-term samples capture local temporal dynamics of a street block, which can then be continuously contrasted along the full time series to identify periods of feature change. Intervals with low semantic similarity are interpreted as changes in street block use types. By analyzing multiple changes over time, the complete construction process of a street block can be derived. In this study, we focus on semantic changes in impervious surfaces as a key land cover type to monitor the construction events within street blocks. For more SSC-SB method details, please refer to <https://github.com/CUG-BEODL/SSC-SB>.

3.1. Semantic encoder training

To enable the semantic encoder to effectively represent short-term street block, a sample set of street blocks with no semantic changes is required, where each sample is paired with corresponding labels for training the encoder. This supervised training approach ensures that the semantic encoder focuses on specific street block use types, thereby minimizing the impact of external noise and enhancing the accuracy and robustness of the model in monitoring tasks. In this study, we adopt a deep learning architecture as the semantic encoder, primarily due to its strong capabilities in automated feature extraction (Krizhevsky et al., 2012).

3.1.1. Stable street blocks labeling

We employed computer vision-based feature extraction methods to efficiently and reliably label street block, following two primary steps: (1) machine annotation using the HOG operators to identify stable street blocks, and (2) manual annotation of street block use types.

- 1. Machine annotation:** The effectiveness of using the HOG operators for extracting street block features has been demonstrated in Wang et al. (2022). We applied it to preliminarily extract the texture features of street blocks. To minimize the impact of seasonal variations on the results, we selected the street block imagery in February of each year from 2021 to 2024 as the input to the HOG operators. A street block was considered a stable sample if the Pearson correlation coefficient between any two of the three time points exceeded 0.7. The calculation formula of Pearson correlation coefficient is as follows:

$$P = \frac{Cov(F_i, F_j)}{\sqrt{Var(F_i) Var(F_j)}} \quad (1)$$

where, F_i and F_j are the feature vectors, and Cov and Var represent covariance and variance, respectively. The similarity value ranges from -1 to 1 .

- 2. Manual annotation:** The imagery of the first time point for each stable street block was visualized for manual labeling. Street block use types were categorized into undeveloped and developed street blocks.

The use types of the stable street blocks selected by the HOG were all manually re-verified, ensuring that no systematic errors were introduced. To ensure the representativeness of the training data while maintaining consistency with the actual distribution of street blocks in each city, we employed a stratified proportional random sampling strategy, using the number of street blocks in each city as the stratification criterion. Specifically, the sample size for each city was allocated

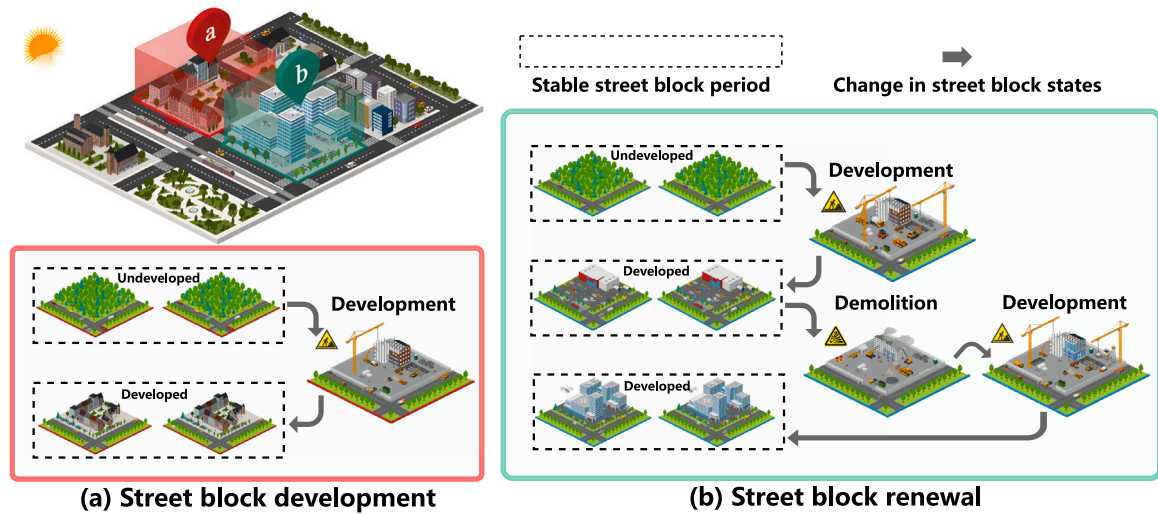


Fig. 2. Two types of trajectories faced by street blocks during urban evolution: (a) street block development, (b) street block renewal.

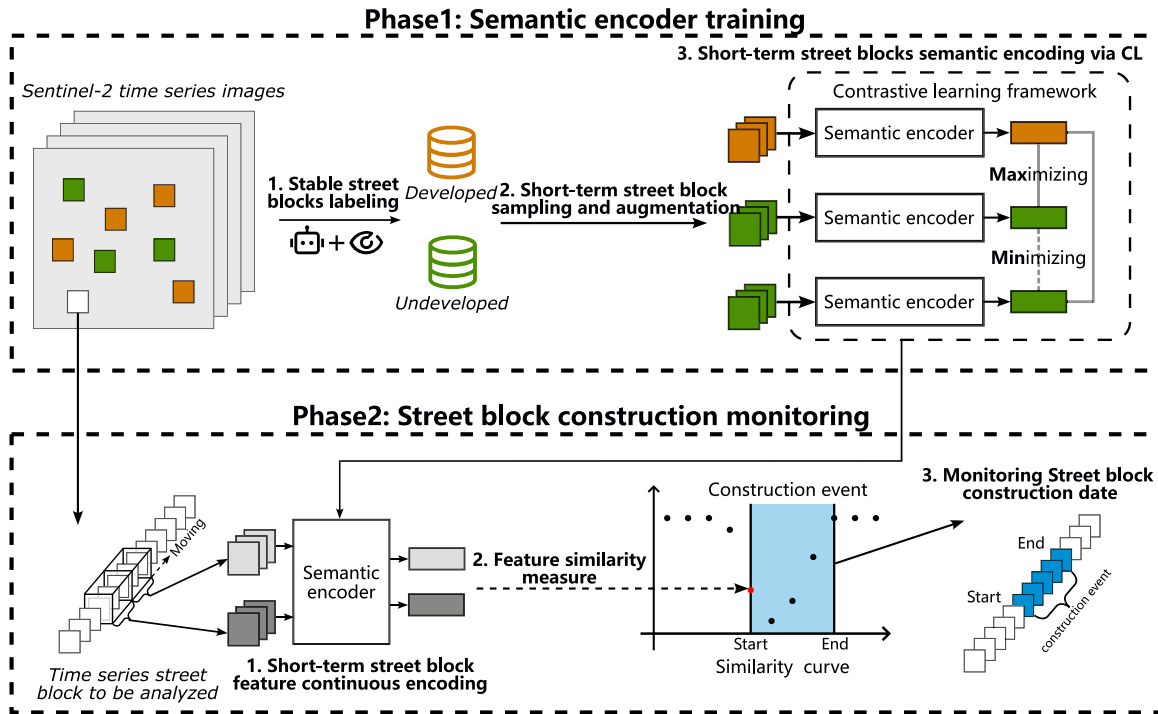


Fig. 3. The flowchart of the proposed SSC-SB.

in proportion to its total number of street blocks. Based on this, approximately 10% of stable street blocks were randomly selected from each city's sample set, yielding a final dataset of 2400 stable street block samples. The ratio of developed to undeveloped samples is about 8:1. Subsequently, 20% of these samples were reserved as the validation set. This approach ensures that the sample distribution aligns with the geographic scale of different cities while mitigating potential sampling bias through random selection, thereby enhancing the model's generalization ability.

3.1.2. Short-term street block sampling and augmentation

We treat a short-term of length *sampleSize* for a street block as a training input sample. The primary goal is to ensure the semantic encoder outputs a feature vector that effectively represents the semantic information of this sample. However, given the limited number of street blocks in a city, constrained by its size and area, the lack of sufficient

samples may hinder the effective training of the semantic encoder. To address this, we applied two data augmentation strategies to expand the training data set:

1. **Random sampling:** For a stable street block with a time series length L , the starting point of the sample is randomly selected within the $[0, L - \text{sampleSize}]$ during each training iteration. This introduces significant randomness to the extracted samples from the same street block across training iterations, helping the model learn more robust features.
2. **Semantic invariance-driven data augmentation:** During training iteration, data augmentation operations such as flipping, rotation, cropping, and adding gaussian noise are applied with a certain probability. These transformations preserve the semantic type of the street block while diversifying the training data.

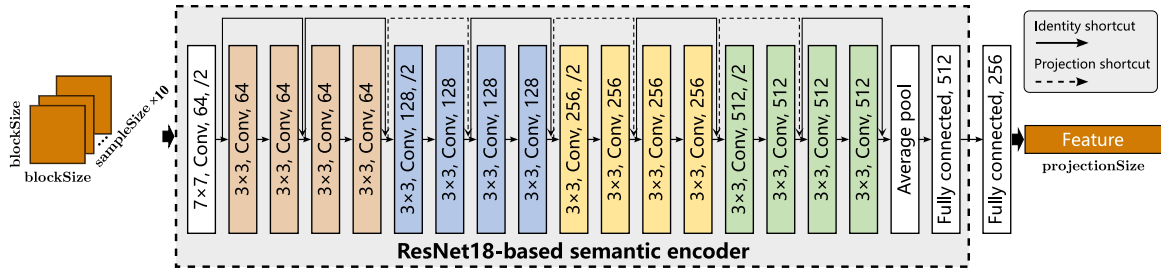


Fig. 4. The network architecture of the semantic encoder based on ResNet18.

3.1.3. Short-term street blocks semantic encoding via contrastive learning

We train the semantic encoder using a supervised contrastive learning framework, which trains the network by maximizing inter-class differences and minimizing intra-class differences (Chen et al., 2020; Deldari et al., 2021). This approach effectively separates the features of developed and undeveloped street blocks, ensuring that street blocks with the same use type have more similar features.

The architecture of the semantic encoder is shown in Fig. 4, where ResNet18 is selected as the backbone network (He et al., 2016). Each short-term street block sample $X \in \mathbb{R}^{sampleSize \times 10 \times W \times H}$ undergoes data dimension alignment, where $sampleSize$ represents the length of the time series, 10 denotes the number of spectral dimensions, and W and H are the width and height of the minimum bounding rectangle of the street block imagery.

1. The temporal dimension $sampleSize$ and spectral dimension (10) of the sample are rearranged into a single dimension, $X_{in} = \{X_1^{b1}, X_1^{b2}, \dots, X_1^{b10}, \dots, X_{sampleSize}^{b1}, X_{sampleSize}^{b2}, \dots, X_{sampleSize}^{b10}\} \in \mathbb{R}^{C \times W \times H}$, where b_i represents the index of spectral band i , and $C = sampleSize \times 10$. We rearrange the spectral and temporal dimensions primarily to reduce computational cost. Without merging, three-dimensional convolution would be required, significantly increasing computational overhead without providing notable improvements in accuracy.
2. Bilinear interpolation is applied to resample the spatial dimensions, constructing the input sample $X_{in} \in \mathbb{R}^{C \times blockSize \times blockSize}$, where $blockSize$ is the size of the resampled street blocks. Spatial resampling is an essential step to align the data, as current network architectures do not support processing inputs with arbitrary dimensions within a single batch.

Chen et al. suggested adding additional fully connected layers after the backbone network to improve feature representation of the data (Chen et al., 2020). Following this recommendation, we added a new fully connected layer after ResNet18 to map the feature dimensions, outputting a feature vector of dimension $projectionSize$ as the semantic representation of the current street block short-term sample. It is worth noting that the backbone network can theoretically be replaced with any model, and since this is not the main focus of this study, we did not conduct a detailed contrast of hyperparameters for the hidden layers, using the default settings (He et al., 2016).

During the training process, two street blocks with the same use type are considered a positive sample pair, while those with different use types are considered a negative sample pair. Therefore, two masks were used to effectively distinguish between positive and negative sample pairs for different street block use types (Deldari et al., 2021): the positive sample mask M_{pos} and the negative sample mask M_{neg} . The positive sample mask is used to identify pairs of samples with the same class label, while the negative sample mask identifies pairs with different class labels. The diagonal elements are excluded from the positive sample mask to avoid including contrast of a sample with itself in the contrastive learning process. The similarity matrix S generated during the contrastive process is shown in Fig. 5, where each batch contains $batchSize$ short-term street block samples. To avoid

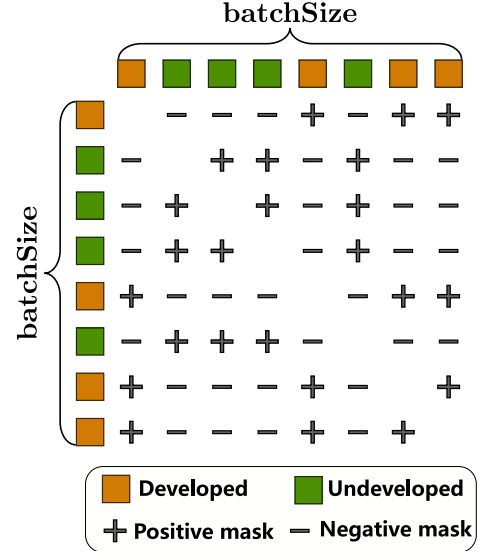


Fig. 5. Diagram of the mask of positive and negative sample pairs in a batch during training.

imbalanced positive and negative sample pairs during training, we ensure that each batch contains an equal number of “developed” and “undeveloped”.

$$M_{pos}(i, j) = \begin{cases} 1, & \text{if } label_i = label_j \text{ and } i \neq j \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

$$M_{neg}(i, j) = \begin{cases} 1, & \text{if } label_i \neq label_j \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

The contrastive loss function is designed to optimize the similarity matrix S , effectively distinguishing between positive and negative sample pairs (Deldari et al., 2021). We refer to the loss function design of supervised contrastive learning (Khosla et al., 2020). For positive sample pairs, the objective is to maximize their similarity, while for negative sample pairs, the goal is to minimize it. The final form of the loss function is implemented using cross-entropy loss, where N represents the number of samples in the batch ($N = batchSize$). We tested three values for the temperature parameter τ (0.1, 0.2, and 0.5) and found minimal impact on model accuracy. We chose $\tau = 0.5$ as it yielded the best performance in our task. We use the Adam optimizer with hyperparameters set to a learning rate of $1e-3$ and weight decay of $1e-6$ to effectively optimize the model parameters.

$$Loss = -\frac{1}{N} \sum_{i=1}^N \log \left(\frac{\sum_{j=1}^N M_{pos}(i, j) \exp(S_{ij}/\tau)}{Y_i} \right) \quad (4)$$

$$Y_i = \sum_{j=1}^N M_{neg}(i, j) \exp(S_{ij}/\tau) + \sum_{j=1}^N M_{pos}(i, j) \exp(S_{ij}/\tau) \quad (5)$$

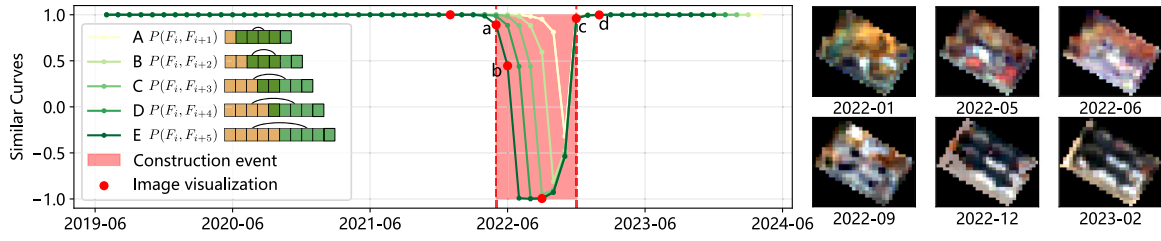


Fig. 6. Similarity curve contrast of street characteristics at different time gaps. The images on the right correspond to the dates of the red dots in the left panel. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2. Street block construction monitoring

3.2.1. Short-term street block feature continuous encoding

For the time series street block imagery of length L , when continuously sampled with a temporal window size of $sampleSize$, each street block can generate $L - sampleSize + 1$ samples. These samples are sequentially input the semantic encoder for semantic representation. Therefore, each street block can produce $L - sampleSize + 1$ feature vectors, each of length $projectionSize$. When a new street block observation is added to the time series, the semantic representation can be further extended by sliding the window forward.

3.2.2. Feature similarity measure

The analysis method based on sequence feature similarity measurement has been widely applied in street block change and disaster detection tasks (Wang et al., 2022; Lê et al., 2022). However, existing methods typically require features from all time points within the time span to detect changes, which does not meet the needs for continuous monitoring of street block construction. In this study, we propose a simple and robust monitoring strategy where the feature vector at each time step is contrasted only with the feature vector from the previous time step, rather than relying on future time step features. This approach enables continuous monitoring of street block construction.

Specifically, we contrast semantic features in chronological order and plot the similarity curves. As shown in Fig. 6, during the feature similarity calculation, we observe that contrasting street block features with a larger time gap results in a more accurate similarity curve. Fig. 6 illustrates several contrast methods using $sampleSize = 5$. When contrasting street block features with a 1-month gap (Fig. 6A), the similarity between the two time periods is relatively high, which may overlook the initial construction event in the street block. In contrast, method E produces the most accurate results, as the two features do not overlap in time, allowing for the fastest response to construction event in the street block. Similarity is evaluated using Pearson's correlation coefficient.

3.2.3. Monitoring street block construction date

We use the following strategies to determine the start and end date of the construction events:

1. When the semantic similarity at time point $date_i$ falls below a threshold $changeThreshold$ (point (a) in Fig. 6), the model triggers a warning for the potential start of construction. When the similarity falls below the threshold again (point (b) in Fig. 6), $date_i$ is confirmed as the start date of street block construction;
2. Similarly, when the semantic similarity at time point $date_j$ returns to or exceeds the threshold $changeThreshold$ (point (c) in Fig. 6), the model records this as a possible construction end date. If the similarity exceeds the threshold again (point (d) in Fig. 6), $date_j$ is confirmed as the construction end date;
3. Finally, based on prior knowledge, we exclude certain false construction events, primarily those where the construction duration is less than 3 months (Ding et al., 2022).

We find that confirming the start and end dates based on two consecutive observations is sufficient to meet the monitoring accuracy requirements, and additional observations do not provide significant improvement. Notably, although this strategy is simple, it demonstrates good robustness in the current task. It relies on sufficiently good feature representations of short-term street block samples to function effectively. This is because high-quality feature representations ensure that the semantic similarity curve is not overwhelmed by excessive noise. Moreover, it eliminates the need for pairwise contrasts of features at each time point (Lê et al., 2022; Wang et al., 2022), enabling continuous monitoring of street block construction. In theory, the SSC-SB method will respond to street block construction after a delay of $sampleSize + 1$ observations.

Table 1 lists several hyperparameters that may be sensitive to numerical values and provides detailed descriptions, along with suggested ranges for reference.

3.3. Street block construction type classification

After identifying the construction events for each street block, two fully connected layers were added after the structure in Fig. 4 to classify the construction events into two types: “development” or “demolition”. The input for the classification model is a time-series sample composed of $sampleSize$ evenly sampled street block images from the identified construction events. If the construction period is shorter than $sampleSize$, additional samples from before the start and after the end of the event are added until the condition is met. We manually labeled 200 street blocks with development events and 200 with demolition events to train the classification model. During training, the semantic encoder shown in Fig. 4 was frozen, and only the parameters of the two fully connected layers were updated. This ensures that the training process focuses on learning to distinguish between different construction types without over-adjusting the learned street block feature representations. This approach preserves the robustness of the classification model and improves training efficiency (Yosinski et al., 2014).

3.4. Accuracy assessment

We labeled 100 street blocks that underwent construction and 100 street blocks with stable use type between January 2021 and December 2024 in Wuhan, Changsha, and Nanchang as the test set. The dataset includes information on construction start and end dates as well as construction types, resulting in a total of 600 samples for evaluating the accuracy of the SSC-SB method. The evaluation metrics are categorized into spatial and temporal domains (He et al., 2024). Each model was trained eight times with different random seeds, and the mean and standard deviation of the evaluation results were reported.

3.4.1. Training stability of semantic encoder

To further evaluate the model's ability to distinguish between positive and negative sample pairs, we introduced the average similarity of positive sample pairs (Sim_{pos}) and negative sample pairs (Sim_{neg}). These two metrics provide an intuitive assessment of the model's ability to differentiate between positive and negative sample pairs,

Table 1

Several potentially sensitive hyperparameters in the SSC-SB method. The bold values indicate the default parameters used in this study.

Hyperparameters	Value	Detail
<i>blockSize</i>	8, 16, 32, 64 , 96	The size of the resampled street blocks
<i>sampleSize</i>	1, 3, 5, 7	Length of the time series for training and inference
<i>batchSize</i>	4, 8, 16, 32, 64 , 128, 256	Batch size in training
<i>projectionSize</i>	8, 16, 32 , 64, 128, 256	Projection output dimensionality
<i>changeThreshold</i>	0.7, 0.8, 0.9 , 0.95, 0.99	Threshold used to detect construction in the similarity curve

Table 2

Confusion matrix for evaluating spatial domain.

Predict	Label	
	Change streets	Stable streets
Change streets	$L_c P_c$	$L_s P_c$
Stable streets	$L_c P_s$	$L_s P_s$

with higher similarity values indicating that the model has successfully identified positive sample pairs, and lower similarity values indicating that negative sample pairs are accurately recognized as dissimilar. The evaluation results for this part can be found in Section 4.1.

$$Sim_{pos} = \frac{\sum_{i,j} M_{pos}(i,j) \cdot S_{ij}}{\sum_{i,j} M_{pos}(i,j)} \quad (6)$$

$$Sim_{neg} = \frac{\sum_{i,j} M_{neg}(i,j) \cdot S_{ij}}{\sum_{i,j} M_{neg}(i,j)} \quad (7)$$

3.4.2. Different semantic encoder training strategies

To evaluate the effectiveness of the SSC-SB under both supervised learning and contrastive learning paradigms, we designed two feature extraction models for comparative experiments. All experiments were conducted under the same hyperparameter settings.

- **Unsupervised + Autoencoder:** An autoencoder is used for feature extraction from the short-term street block sample. Specifically, a deconvolution layer is added after the average pooling layer in Fig. 4, with two deconvolution layers used to restore the image size and spectral channels. The mean squared error is adopted as the loss function.
- **Supervised + Classification:** The contrastive learning framework is replaced with a supervised classification task, while keeping the other module structures unchanged. A fully connected layer is appended at the end to output two classes, undeveloped and developed. This approach is trained with labeled data, helping the model learn clear class information, thus enabling more precise feature classification.

3.4.3. Spatial domain metrics

The spatial domain evaluation mainly assesses the model's accuracy in determining whether a street block has undergone construction, which can be considered as a binary classification task: stable street block vs. change street block. The confusion matrix for spatial domain evaluation is shown in Table 2. The study primarily uses the S_{OA} metric to quantify spatial accuracy, calculated as follows:

$$S_{OA} = \frac{L_c P_c + L_s P_s}{L_c P_c + L_s P_s + L_c P_s + L_s P_c} \times 100\% \quad (8)$$

3.4.4. Temporal domain metrics

Three evaluation metrics are used to assess the model's accuracy in detecting street block construction temporal span. First, we use the T_{start} and T_{end} metrics to evaluate the accuracy of the predicted start and end dates of construction for street blocks:

$$T_{start} = \frac{\#[\sum_i^N \sum_j^{K_i} (L_{start_j}^i == P_{start_j}^i)]}{\sum_i^N K_i} \times 100\% \quad (9)$$

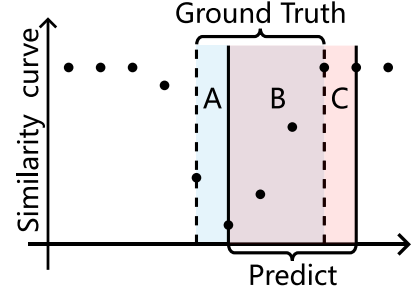


Fig. 7. Schematic diagram for calculating T_{mIoU} .

Table 3

Confusion matrix for evaluating construction event classification.

Predict	Label	
	Development	Demolition
Development	$L_{dev} P_{dev}$	$L_{dem} P_{dev}$
Demolition	$L_{dev} P_{dem}$	$L_{dem} P_{dem}$

$$T_{end} = \frac{\#[\sum_i^N \sum_j^{K_i} (L_{end_j}^i == P_{end_j}^i)]}{\sum_i^N K_i} \times 100\% \quad (10)$$

where N represents the number of construction street blocks in the test dataset that are correctly detected by SSC-SB, which are used for temporal domain metrics (Zhu and Woodcock, 2014), K_i is the number of construction events for the i th street block, and $\#$ is the counting function, representing the number of times the predicted start date matches the true start date, or the predicted end date matches the true end date. To account for potential labeling errors, we allow a small margin of error for the two months (He et al., 2024; Xu et al., 2019a). Additionally, we use the T_{mIoU} metric to evaluate the model's ability to coverage the construction period of street blocks. As shown in Fig. 7, the period ($A + B$) represents the true construction temporal span of the street block, while ($B + C$) represents the predicted construction temporal span. The intersection-over-union (IoU) is calculated as ($Iou = B / (A + B + C)$). A higher IoU value indicates better detection accuracy. We compute this metric for each construction event of every street block and take the average as the overall detection accuracy.

$$T_{mIoU} = \frac{\sum_i^N \sum_j^{K_i} B_j^i}{\sum_i^N \sum_j^{K_i} (A_j^i + B_j^i + C_j^i)} \times 100\% \quad (11)$$

3.4.5. Classification metrics

The confusion matrix for evaluating construction event classification is shown in Table 3. Overall classification accuracy (C_{OA}), a widely used metric in classification tasks, is employed in this study to assess the model's capability in classifying street block construction event.

$$C_{OA} = \frac{L_{dev} P_{dev} + L_{dem} P_{dem}}{L_{dev} P_{dev} + L_{dem} P_{dem} + L_{dem} P_{dev} + L_{dev} P_{dem}} \times 100\% \quad (12)$$

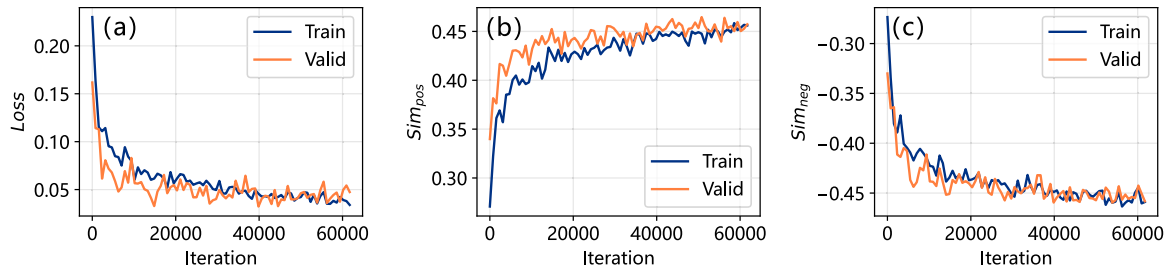


Fig. 8. The changing curves of several indicators of the training set and the validation set during the training process, (a) loss change, (b) similarity change between positive samples and (c) similarity change between negative samples.

Table 4

Accuracy assessment results of the proposed method.

City	S_{OA} (%)	T_{start} (%)	T_{end} (%)	T_{mIou} (%)	C_{OA} (%)
Wuhan	87.8	65.9	52.1	64.8	85.2
Changsha	92.1	72.3	59.4	67.2	92.4
Nanchang	91.3	68.2	53.2	67.5	90.7
Average	90.4 ± 3.2	68.8 ± 3.0	54.9 ± 1.8	66.5 ± 2.7	89.4 ± 1.1

4. Results

4.1. Semantic encoder training process

Fig. 8 visualizes the contrastive training process of the semantic encoder with a batch size of 64. From panel (a), it is evident that the contrastive loss significantly decreases on both the training and validation datasets. Panels (b) and (c) show the differences in similarity metrics between positive and negative sample pairs. It can be observed that the semantic encoder gradually increases the similarity within positive sample pairs while decreasing the similarity within negative sample pairs. The model's convergence demonstrates the feasibility of using contrastive learning with positive and negative sample pairs for semantic representation of street blocks.

4.2. Monitor street blocks construction accuracy

Table 4 presents the accuracy of street block development and renewal in the three representative cities using the proposed method, with all cities achieving satisfactory results. Notably, the model exhibits strong capability in determining whether a street block has undergone change, achieving an overall detection accuracy of 90.4%. We attribute this to the extraction of deeper semantic features, which effectively reduces false changes and enhances the robustness of change detection. The method's ability to capture the temporal span of street block construction, measured by T_{mIou} , averages 66.5%, indicating that the proposed approach can detect most street block construction events with high accuracy. For each detected construction, the method demonstrates better performance in identifying the start time of construction compared to the end time. This is likely because construction typically begins with rapid and significant land cover changes over a short period, whereas later stages involve more gradual and subtle transformations, making detection more challenging. Nevertheless, the model delivers reliable monitoring results overall. The classification accuracy of street block construction achieves an average of 89.4%, which is generally sufficient for urban renewal analysis.

Fig. 9 shows the accuracy evaluation results of the three methods. Compared to the unsupervised feature extraction method, the supervised method helps the model learn meaningful semantic change information in the street block construction monitoring task more effectively, usually yielding better accuracy. This result demonstrates that supervised learning is particularly advantageous for monitoring tasks involving complex and diverse change patterns. Regarding contrastive

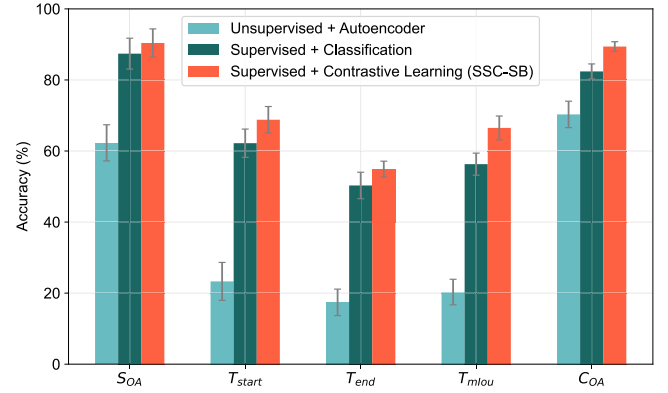


Fig. 9. Accuracy comparison of feature extraction models: Unsupervised + Autoencoder, Supervised + Classification, and Supervised + Contrastive Learning. The figure reports both the mean accuracy and the standard deviation for each method.

learning, the results further indicate that this paradigm outperforms traditional supervised classification in terms of accuracy. We believe this is due to contrastive learning's ability to optimize the distance between similar and dissimilar samples, effectively capturing the latent semantic structure within the data, thus enhancing the model's ability to distinguish complex change patterns. Consequently, contrastive learning can deliver higher accuracy than supervised classification.

4.3. Comparison to other block-level change detection studies

Compared to prior studies that inspired our work, including MTS-GS (Wang et al., 2022), which also utilizes similarity-based approaches for block-level change detection, our proposed method is more specifically designed to detect construction-related events during street block development and renewal. In addition, we also include the TSFLC method, which detects street block changes by analyzing temporal variations in the fractional composition of land cover (Jing et al., 2021). These two methods represent distinct paradigms in block-level change detection and are included as baselines for comparative analysis. Here, we evaluate their performance specifically in terms of detecting the start and end of changes. It is important to note that neither of these methods was originally designed to address the specific objectives of our study. Therefore, while they serve as the closest available references, the comparisons are not entirely equivalent. The reported results are intended primarily for qualitative reference, rather than definitive benchmarking.

Table 5 presents the accuracy evaluation results of the three methods. Since the implementations of MTS-GS and TSFLC are not fully publicly available, we reproduced both methods based on the descriptions provided in the original papers and performed quantitative evaluation through manual annotation. The monthly land cover maps required by TSFLC were generated using the TSSCD method (He et al.,

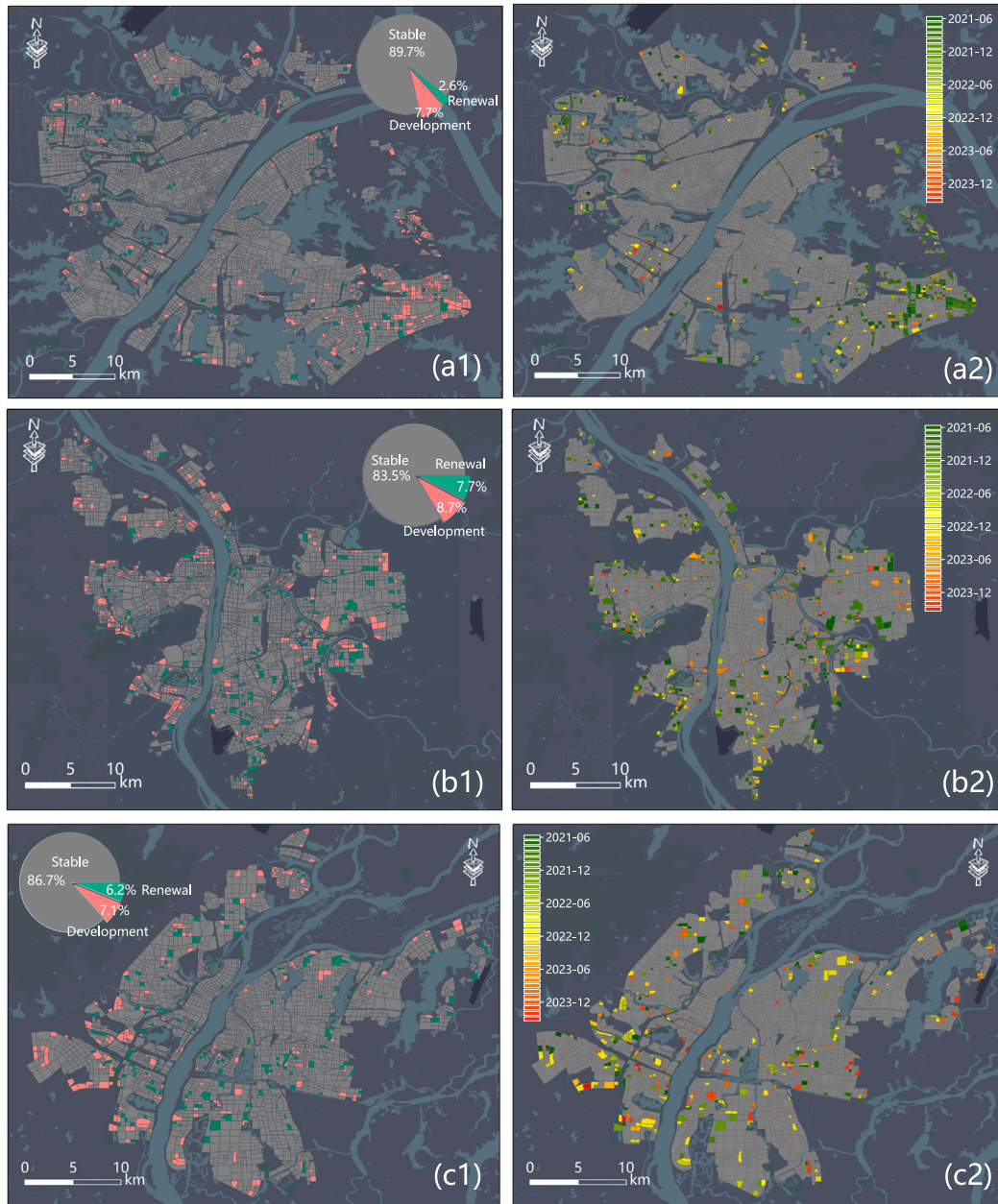


Fig. 10. Inference results for Wuhan, Changsha, and Nanchang using the proposed method. The left panel presents the distribution and proportion of different street block change types, while the right panel illustrates the timing of the first construction activity for each street block.

2024). The results show that SSC-SB significantly outperforms MTS-GS, an unsupervised method. Further analysis indicates that the HOG-based used in MTS-GS lacks stability when applied to small-scale street blocks. This issue is particularly evident in our study area, which features hilly terrain and fragmented block boundaries, leading to degraded feature matching and reduced detection accuracy. In contrast, the TSFLC method achieves higher accuracy than SSC-SB in detecting construction temporal span (T_{start} , T_{end} , and T_{mIou}). This approach is based on pixel-level changes in land cover proportions, which can intuitively reveal the evolution of street block use types and provides certain advantages in detection. However, its accuracy in determining whether a street block has undergone changes (S_{OA}) is lower than that of SSC-SB. Upon inspection, the errors of TSFLC are mainly concentrated in samples where multiple land cover types coexist within a block. In such cases, the proportion of impervious surfaces fluctuates over time, leading to

Table 5

Comparison of the accuracy of block-level change detection methods.

Method	S_{OA} (%)	T_{start} (%)	T_{end} (%)	T_{mIou} (%)
MTS-GS	62.7	23.3	25.4	29.3
TSFLC	78.5	71.2	60.8	67.2
SSC-SB	90.4	68.8	54.9	66.5

the detection of numerous false change street blocks. Overall, SSC-SB achieves a favorable balance between accuracy, adaptability, and computational efficiency. It performs reliably in complex urban terrains and demonstrates strong scalability, making it suitable for large-scale, long-term monitoring of street block changes in the context of urban renewal.

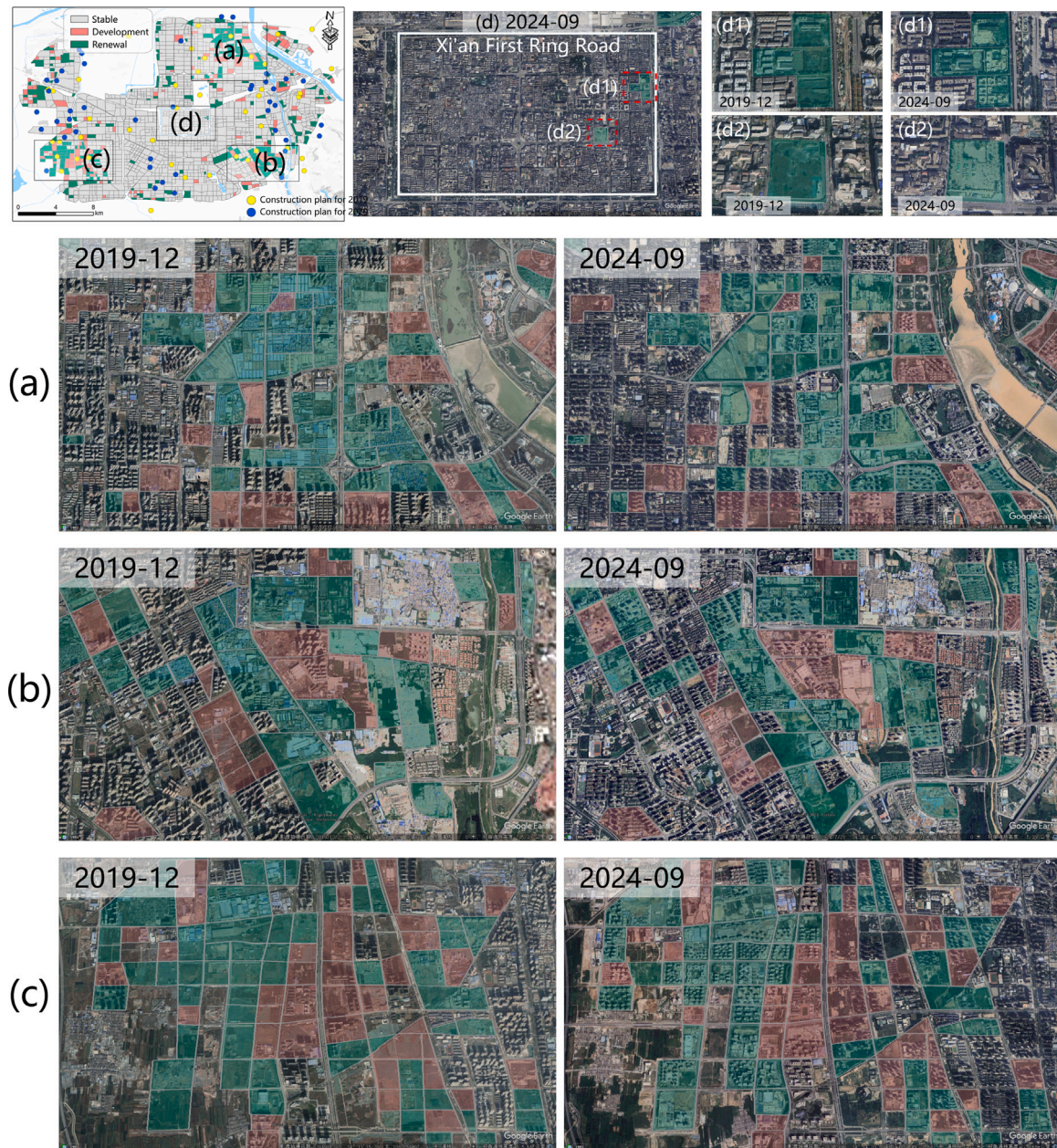


Fig. 11. High-resolution Google Earth images from December 2019 and September 2024, showing three areas in Xi'an with concentrated construction events and a region within the first ring road of the urban core with minimal street block changes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.4. Qualitative visualization of block construction monitoring

We applied the trained SSC-SB to infer street blocks across the MYRB, obtaining results on street block construction from 2021 to 2024. Fig. 10 presents the results for Wuhan, Changsha, and Nanchang. The left panel illustrates the types of street block changes from 2021 to 2024. Visually, street block development is concentrated on the urban periphery, reflecting urban expansion, which is particularly evident in Wuhan, where the eastern region has experienced rapid growth over the past four years. In contrast, street block renewal is primarily distributed within the urban core.

To further demonstrate the effectiveness of the proposed method, we additionally analyzed Xi'an, one of the first pilot cities for urban renewal, which has undergone significant urban renewal in recent years. The analysis is based on data from January 2019 to December 2024. We examined the urban development targets set by the local government

in 2018, which aimed to clear and renew 105 urban villages between 2019 and 2020. This included complete village demolition and waste removal. As shown in the top-right subfigure of Fig. 11, yellow dots represent the 60 urban villages planned for 2019, while blue dots denote the 45 villages planned for 2020. The SSC-SB method's detection results show a notable overlap with the planned developments, further validating its effectiveness. Notably, due to the impact of the COVID-19 pandemic and policy adjustments, not all planned street blocks progressed as expected. By examining high-resolution Google Earth imagery from December 2019 and September 2024, we identified three regions in Xi'an with concentrated construction activity, as depicted in Fig. 11(a)–(c). The SSC-SB method effectively distinguished between stable street blocks and those that underwent constructions. As shown in Fig. 11(d), only four street blocks in the urban core (within the first ring road) were identified as undergoing renewal. This result further

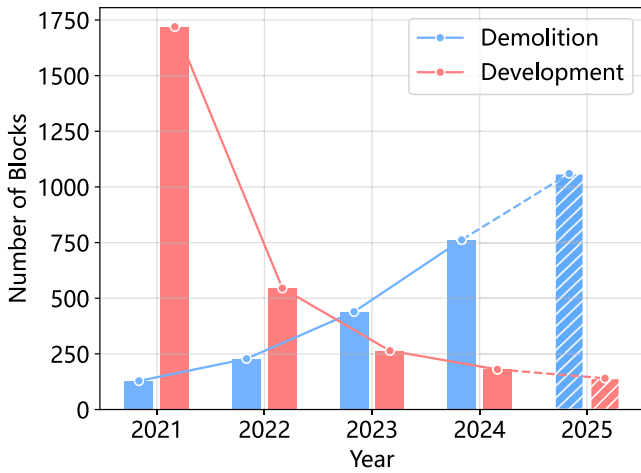


Fig. 12. The annual statistics of street blocks undergoing construction from 2021 to 2024.

demonstrates the reliability of the SSC-SB method, as it avoids over-detection of changed street blocks that could mislead users. Instead, SSC-SB provides accurate and realistic results. This capability offers valuable insights for governments to assess urban renewal progress from a macro perspective.

4.5. MYRB urban development and renewal statistics results

Fig. 12 illustrates the annual trends in street block construction from 2021 to 2024. Since both 2021 and 2024 require a certain period for model initialization and inference, the annual statistics for these two years are replaced by multiplying the average monthly number of construction street blocks by 12. The results reveal a distinct decline in the number of developed street blocks, while the number of demolished street blocks has been increasing sharply. Notably, 2023 represents a pivotal year, as the number of demolished street blocks exceeded that of developed ones for the first time. A straightforward projection suggests that by 2025, over 1000 street blocks will undergo demolition, reflecting a shift toward urban renewal, whereas the number of developed street blocks may decrease to approximately 200.

Fig. 13 illustrates the distribution of street block construction types across 30 cities in the MYRB region, where the proportion of changed street blocks ranges from 5% to 20%. Ezhou, Changsha, Changde, and Pingxiang exhibit relatively high proportions of renewal. In particular, Ezhou ranks first in Hubei Province in terms of renewal share, indicating active reconstruction or functional transformation of existing street blocks. As representative cities of Hunan Province, Changsha and Changde also show strong renewal trends, which may reflect efforts to optimize internal urban structure following expansion. Although Pingxiang has a lower overall construction ratio, the high share of renewal suggests a development pattern primarily driven by old town renovation. To assess urban construction dynamics, we employ two indicators: the street block construction rate ($R_{con.}$) and the renewal rate ($R_{ren.}$), defined as follows:

$$R_{con.} = \frac{N_d + N_r}{N_d + N_r + N_s} \times 100\% \quad (13)$$

$$R_{ren.} = \frac{N_r}{N_d + N_r} \times 100\% \quad (14)$$

where N_d , N_r , and N_s denote the number of developed, renewed, and stable street blocks, respectively. As shown in Table 6, Hunan Province exhibits the highest $R_{con.}$ and $R_{ren.}$ values, indicating the most active urban construction. The relatively high proportion of renewal suggests that Hunan is not only expanding new urban areas but also

Table 6

The street block construction rate ($R_{con.}$) and the renewal rate ($R_{ren.}$) across different provinces.

Province	N_d	N_r	$R_{con.}$	$R_{ren.}$
Hubei	739	304	9.7%	29.1%
Hunan	553	393	13.1%	41.5%
Jiangxi	456	266	12.0%	36.8%

prioritizing the upgrading of existing ones. In contrast, Hubei Province reports the lowest $R_{con.}$ and $R_{ren.}$, implying weaker overall construction activity, primarily driven by new development, with limited investment in upgrading existing street blocks. This suggests that urban growth in Hubei remains in an outward expansion phase, with slower progress in urban renewal. Jiangxi Province falls between Hunan and Hubei in both $R_{con.}$ and $R_{ren.}$.

Fig. 14 illustrates the distribution of street block development and demolition durations. The results indicate that, on average, construction in the MYRB is completed within 6–8 months. Notably, the duration of development exhibits significantly less variability than that of demolition. This discrepancy may stem from the fact that development projects are typically well-planned and systematically managed, resulting in more consistent timelines with minimal fluctuation. In contrast, demolition timelines are subject to various policy, social, and economic influences, leading to greater variability across cities and regions.

5. Discussion

5.1. Qualitative analysis of semantic encoding advantages

We selected several representative cases of street block change and stable cases for visualization to qualitatively assess the advantages of semantic features over shallow features, such as texture. Fig. 15(a)–(c) depict three distinct construction events: development, demolition, and development following demolition. The green lines represent similarity curves derived from texture features of single-temporal street blocks, extracted using HOG operators. The results indicate that similarity curves based on semantic features exhibit stable trends with minimal noise, effectively capturing changes in street block use types. When construction event occurs, the curve displays a distinct “U” shape. Fig. 15(d) and (e) present cases of developed and undeveloped street blocks where no construction activity has taken place. As expected, the similarity curve remains stable due to the absence of detectable changes in street block use type. However, texture-based similarity, as a shallow feature, fails to effectively represent the construction process. First, it is highly sensitive to extraneous texture variations, such as those caused by cloud cover or seasonal vegetation changes, as observed in (d) and (e), which are unrelated to human interventions. Second, it struggles to directly capture actual changes in street block use types. For instance, despite substantial development and demolition in (a) and (b), texture-based similarity fails to accurately distinguish these construction events

5.2. Model transferability assessment

We evaluated the transferability of SSC-SB using Xi'an from January 2019 to December 2024 as the target domain. As a capital province in northern China, Xi'an is characterized by relatively flat terrain and regularly shaped street blocks, which stands in sharp contrast to the complex topography and irregular block patterns of cities in the MYRB. This distinction provides a representative case for assessing the model's ability to generalize and adapt across different urban forms. The source domain models are as follows:

- SSC-SB_{WH}, was trained using samples from Wuhan.

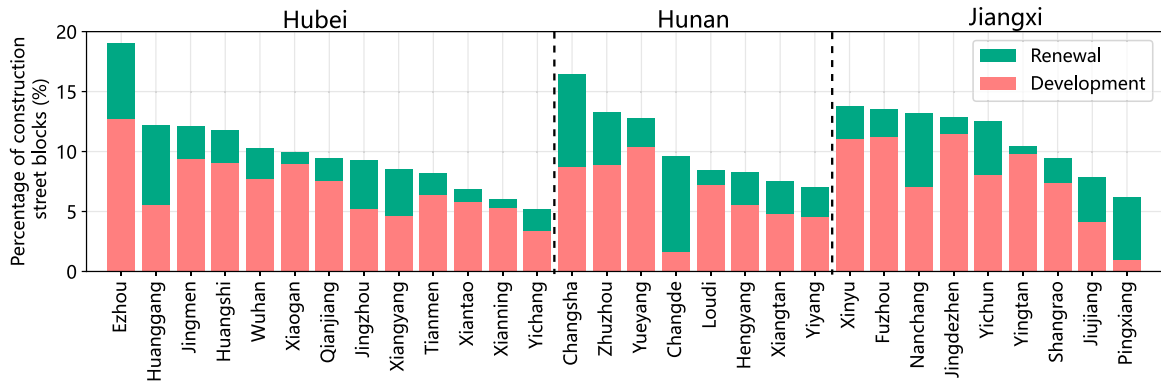


Fig. 13. The distribution of street block construction types across 30 cities in the MYRB region.

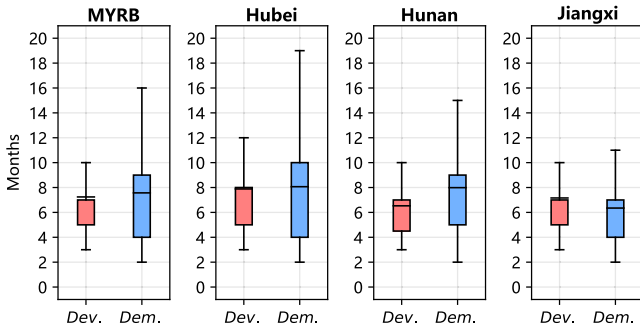


Fig. 14. The distribution of street block development and demolition durations.

- SSC-SB_{MYRB}, was trained on samples from MYRB, with and without data augmentation.
- SSC-SB_{XA}, was trained using 200 labeled samples from Xi'an.

As shown in Fig. 16, SSC-SB_{WH} yielded the lowest performance across all four metrics, indicating that a model trained on a single-city dataset struggles to generalize when transferred to a different urban context. In contrast, SSC-SB_{MYRB}, trained on a broader and more diverse set of street block samples from the MYRB region, significantly improved transfer performance. With data augmentation, it achieved up to 80% of the performance of SSC-SB_{XA} on four metrics. These results demonstrate the model's strong cross-domain adaptability and its ability to capture shared patterns of urban development across cities. Nevertheless, SSC-SB_{XA}, trained on a small number of target-domain samples, still achieved the best overall performance, suggesting that local training remains the most effective approach when limited annotated data are available in the target area. In summary, SSC-SB exhibits promising transferability. With sufficient training data and proper augmentation strategies, it can achieve near-local performance even without access to target city samples, offering a viable solution for large-scale urban construction monitoring.

5.3. Hyperparameter sensitivity analysis

We performed a sensitivity analysis on the six hyperparameters involved in the SSC-SB method, as shown in supplementary material Figure S1. For each experiment, the hyperparameters that remained unchanged were set according to the default values. The parameters *blockSize*, *sampleSize*, and *batchSize* exhibited a certain level of sensitivity. In contrast, the parameters *projectionSize* and *changeThreshold* showed little sensitivity to changes in their values. Based on this, we provide the following analysis:

We conducted a sensitivity analysis on the six hyperparameters involved in the SSC-SB method, as presented in Supplementary Material Figure S1. In each experiment, the unchanged hyperparameters were set to their default values. The parameters *blockSize*, *sampleSize*, and *batchSize* exhibited a certain degree of sensitivity, whereas *projectionSize* and *changeThreshold* showed minimal sensitivity to variations in their values. Based on these findings, we provide the following analysis:

- *blockSize*: As illustrated in Figure S1(a), increasing the input street block size preserves more detailed information, leading to higher accuracy.
- *sampleSize*: As illustrated in Figure S1(b), accuracy initially increases and then declines as the short-term street block sample length increases. When *sampleSize* = 1, the sample fails to effectively capture street block features across months, making it highly susceptible to noise, such as cloud cover. Conversely, when *sampleSize* = 7, the sample may span the entire construction process of the street block, reducing sensitivity to changes.
- *batchSize*: As illustrated in Figure S1(c), as *batchSize* increases, more positive and negative sample pairs can be contrasted within each batch, providing the semantic encoder with additional information to optimize its parameters. However, a larger *batchSize* does not necessarily yield better results.

5.4. Street view images for validating SSC-SB

Remote sensing imagery primarily provides a top-down perspective, which limits its ability to fully capture street block construction events and may introduce uncertainties in the monitoring process. As illustrated in Fig. 17, we utilize Baidu Maps street view images to assess these uncertainties in street block construction detection using the SSC-SB method. The figure presents a specific case in which SSC-SB detection results indicate that the street block was in the development event from August 2021 to August 2022. However, street view images reveal that construction was still ongoing as of September 2022. Additionally, the right side of Fig. 17 displays Sentinel-2 imagery from throughout 2022. The observed uncertainties can be primarily attributed to the inherent limitations of satellite remote sensing data. On one hand, once a building's foundation is completed, the land cover type within the street block undergoes a significant change, leading to a classification shift in street block use. On the other hand, as building heights increase, variations in land cover and texture features become less distinguishable, making it challenging to accurately determine the timing of changes using remote sensing imagery. This issue becomes particularly pronounced after August 2022, when changes within the street block become increasingly difficult to detect. This phenomenon not only underscores the limitations of the SSC-SB method but also highlights broader challenges faced by street block change detection

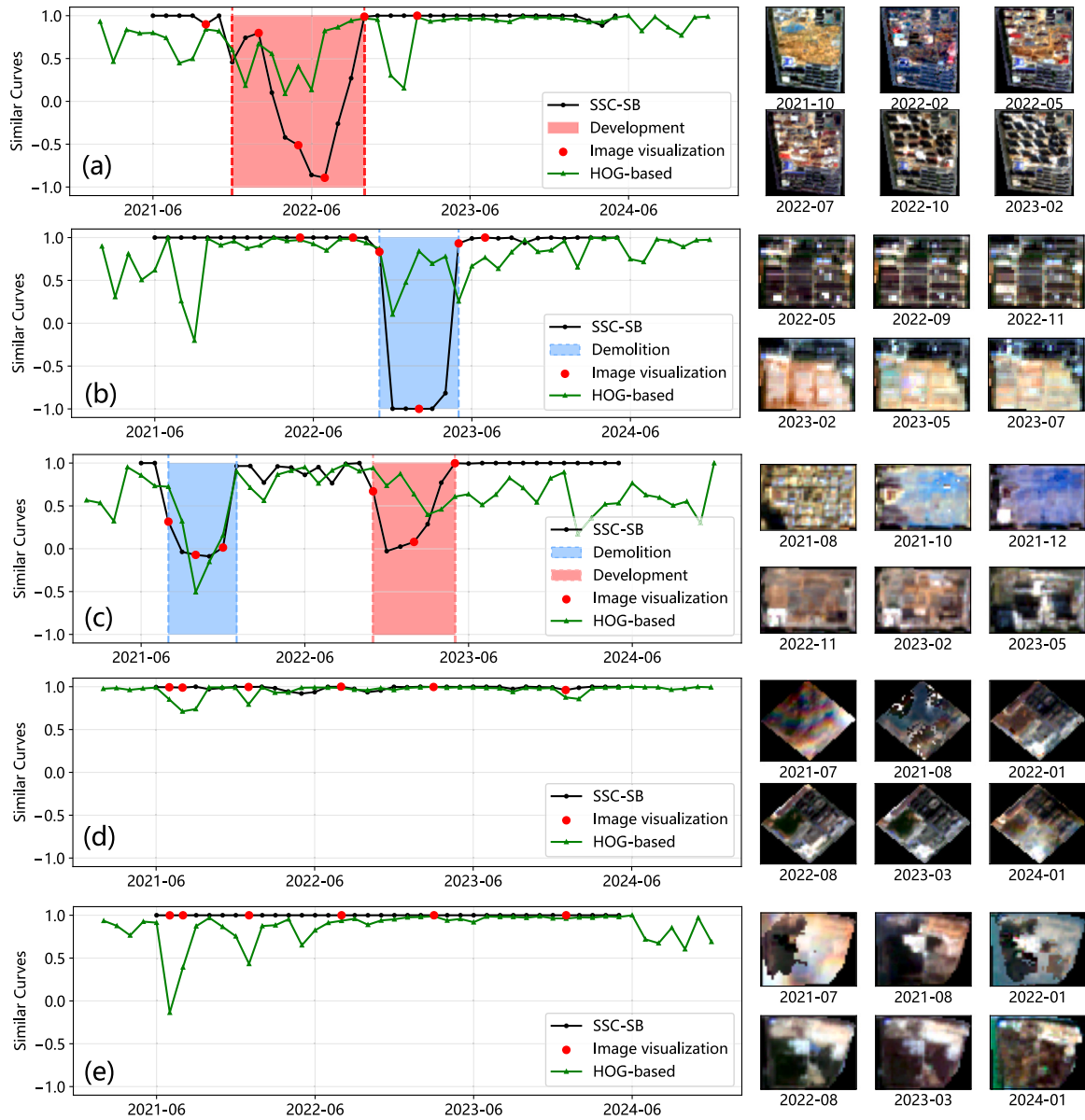


Fig. 15. Similarity curves of several different street blocks construction types or stable street blocks obtained based on the SCC-SB and HOG. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

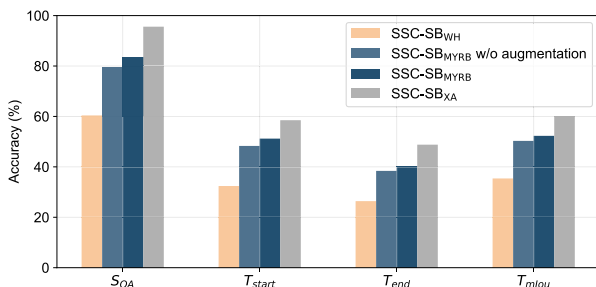


Fig. 16. Comparison of street development and renewal monitoring accuracy in the target domain (Xi'an) under different source domain training settings.

models that rely solely on single-source remote sensing data (Jing et al., 2021; Wang et al., 2022). Future research could explore the integration of multi-source data, such as higher-resolution imagery, nighttime light

data, or social sensing data, to overcome the constraints of single-perspective remote sensing imagery and enhance the accuracy and reliability of street block construction monitoring.

6. Conclusions

This study proposed a semantic contrastive learning framework for monitoring street block dynamics using time-series remote sensing imagery. Applied to the Middle Yangtze River Basin (MYRB) urban agglomeration, covering 30 cities and more than 24,000 street blocks, the framework achieved high accuracy in detecting both development and renewal processes, confirming its scalability and practical applicability at a regional scale. The key contribution of this work lies in introducing a novel monitoring paradigm that integrates semantic encoding with temporal contrast, enhancing sensitivity to functional changes while reducing noise. This approach enables fine-grained tracking of urban development and renewal and demonstrates promising transferability to other rapidly evolving urban environments.



Fig. 17. Verifying the accuracy of street block construction monitoring from a street view perspective.

Despite the promising results, certain limitations remain. The framework depends on the quality and temporal resolution of available imagery, which may reduce robustness in regions with sparse observations or persistent cloud cover. Furthermore, while the method effectively captures construction dynamics, integrating socioeconomic and policy-related data would allow for a more comprehensive interpretation of urban transformation. Future research will focus on extending the framework to multi-source data fusion, incorporating higher-resolution and multimodal imagery to improve detection accuracy.

CRedit authorship contribution statement

Haixu He: Writing – original draft, Validation, Supervision, Methodology, Data curation. **Jining Yan:** Validation, Supervision, Funding acquisition. **Lirong Liu:** Methodology, Formal analysis. **Xu Long:** Writing – original draft, Methodology. **Runyu Fan:** Formal analysis, Data curation. **Zhongchang Sun:** Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.rse.2025.115070>.

Data availability

Data will be made available on request.

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